Hierarchical Linear Models

Applications and Data Analysis Methods

Second Edition

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Series Editor's Introduction to Hierarchical Linear Models

In the social sciences, data structures are often hierarchical in the following sense: We have variables describing individuals, but the individuals also are grouped into larger units, each unit consisting of a number of individuals. We also have variables describing these higher order units.

The leading example is, perhaps, in education. Students are grouped in classes. We have variables describing students and variables describing classes. It is possible that the variables describing classes are aggregated student variables, such as number of students or average socioeconomic status. But the class variables could also describe the teacher (if the class has only one teacher) or the classroom (if the class always meets in the same room). Moreover, in this particular example, further hierarchical structure often occurs quite naturally. Classes are grouped in schools, schools in school districts, and so on. We may have variables describing school districts and variables describing schools (teaching style, school building, neighborhood, and so on).

Once we have discovered this one example of a hierarchical data structure, we see many of them. They occur naturally in geography and (regional) economics. In a sense, one of the basic problems of sociology is to relate properties of individuals and properties of groups and structures in which the individuals function. In the same way, in economics there is the problem of relating the micro and the macro levels. Moreover, many repeated measurements are hierarchical. If we follow individuals over time, then the measurements for any particular individual are a group, in the same way as the school class is a group. If each interviewer interviews a group of interviewees, then the interviewers are the higher level. Thinking about these hierarchical structures a bit longer inevitably leads to the conclusion that many, if not most, social science data have this nested or hierarchical structure.

The next step, after realizing how important hierarchical data are, is to think of ways in which statistical techniques should take this hierarchical structure into account. There are two obvious procedures that have been somewhat discredited. The first is to disaggregate all higher order variables to the individual level. Teacher, class, and school characteristics are all assigned to the individual, and the analysis is done on the individual level. The problem with this approach is that if we know that students are in the same class, then we also know that they have the same value on each of the class variables. Thus we cannot use the assumption of independence of observations that is basic for the classical statistical techniques. The other alternative is to aggregate the individual-level variables to the higher level and do the analysis on the higher level. Thus we aggregate student characteristics over classes and do a class analysis, perhaps weighted with class size. The main problem here is that we throw away all the within-group information, which may be as much as 80% or 90% of the total variation before we start the analysis. As a consequence, relations between aggregated variables are often much stronger, and they can be very different from the relation between the nonaggregate variables. Thus we waste information, and we distort interpretation if we try to interpret the aggregate analysis on the individual level. Thus aggregating and disaggregating are both unsatisfactory.

If we limit ourselves to traditional linear model analysis, we know that the basic assumptions are linearity, normality, homoscedasticity, and independence. We would like to maintain the first two, but the last two (especially the independence assumption) should be adapted. The general idea behind such adaptations is that individuals in the same group are closer or more similiar than individuals in different groups. Thus students in different classes can be independent, but students in the same class share values on many more variables. Some of these variables will not be observed, which means that they vanish into the error term of the linear model, causing correlation between disturbances. This idea can be formalized by using variance component models. The disturbances have a group and an individual component. Individual components are all independent; group components are independent between groups but perfectly correlated within groups. Some groups might be more homogeneous than other groups, which means that the variance of the group components can differ.

There is a slightly different way to formalize this idea. We can suppose that each of the groups has a different regression model, in the simple regression case with its own intercept and its own slope. Because groups are also sampled, we then can make the assumption that the intercepts and slopes are a random sample from a population of group intercepts and slopes. This defines random-coefficient regression models. If we assume this for the intercepts only, and we let all slopes be the same, we are in the variance-component situation discussed in the previous paragraph. If the slopes vary randomly as well, we have a more complicated class of models in which the covariances of the disturbances depend on the values of the individual-level predictors.

In random-coefficient regression models, there is still no possibility to incorporate higher level variables, describing classes or schools. For this we need multilevel models, in which the group-level model is again a linear model. Thus we assume that the slope of the student variable SAT depends linearly on the class variables of class size or teacher philosophy. There are linear models on both levels, and if there are more levels, there are more nested linear models. Thus we arrive at a class of models that takes hierarchical structure into account and that makes it possible to incorporate variables from all levels.

Until about 10 years ago, fitting such models was technically not possible. Then, roughly at the same time, techniques and computer programs were published by Aitkin and Longford, Goldstein and co-workers, and Raudenbush and Bryk. The program HLM, by Bryk and Raudenbush, was the friendliest and most polished of these products, and in rapid succession a number of convincing and interesting examples were published. In this book, Bryk and Raudenbush describe the model, the algorithm, the program, and the examples in great detail. I think such a complete treatment of this class of techniques is both important and timely. Hierarchical linear models, or multilevel models, are certainly not a solution to all the data analysis problems of the social sciences. For this they are far too limited, because they are still based on the assumptions of linearity and normality, and because they still study the relatively simple regression structure in which a single variable depends on a number of others. Nevertheless, technically they are a big step ahead of the aggregation and disaggregation methods, mainly because they are statistically correct and do not waste information.

I think the main gain, illustrated nicely in this book by the extensive analysis of the examples, is conceptual. The models for the various levels are nicely separated, without being completely disjointed. One can think about the possible mechanisms on each of the levels separately and then join the separate models in a joint analysis. In educational research, as well as in geography, sociology, and economics, these techniques will gain in importance in the next few years, until they also run into their natural limitations. To avoid these limitations, they will be extended (and have been extended) to more levels, multivariate data, path-analysis models, latent variables, nominal-dependent variables, generalized linear models, and so on. Social statisticians will be able to do more extensive modeling, and they will be able to choose from a much larger class of models. If they are able to build up the necessary prior information to make a rational choice from the model class, then they can expect more power and precision. It is a good idea to keep this in the back of your mind as you use this book to explore this new exciting class of techniques.

JAN DE LEEUW
SERIES EDITOR

1 Introduction

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Hierarchical Data Structure: A Common Phenomenon

Much social research involves hierarchical data structures. In organizational studies, researchers might investigate how workplace characteristics, such as centralization of decision making, influence worker productivity. Both workers and firms are units in the analysis; variables are measured at both levels. Such data have a hierarchical structure with individual workers nested within firms.

In cross-national studies, demographers examine how differences in national economic development interact with adult educational attainment to influence fertility rates (see, e.g., Mason, Wong, & Entwistle, 1983). Such research combines economic indicators collected at the national level with household information on education and fertility. Both households and countries are units in the research with households nested within countries, and the basic data structure is again hierarchical.

Similar kinds of data occur in developmental research where multiple observations are gathered over time on a set of persons. The repeated measures contain information about each individual's growth trajectory. The psychologist is typically interested in how characteristics of the person, including variations in environmental exposure, influence the shape of these growth trajectories. For example, Huttenlocher, Haight, Bryk, and Seltzer (1991) investigated how differences among children in exposure to language in the home predicted the development of each child's vocabulary over time. When every person is observed at the same fixed number of time points, it is conventional to view the design as occasions crossed by persons. But when the number and spacing of time points vary from person to person, we may view occasions as nested within persons.

The quantitative synthesis of findings from many studies presents another hierarchical data problem. An investigator may wish to discover how differences between studies in treatment implementations, research methods, subject characteristics, and contexts relate to treatment effect estimates within studies. In this case, subjects are nested within studies. Hierarchical models provide a very general statistical framework for these research activities (Berkey et al., 1995; Morris & Normand, 1992; Raudenbush & Bryk, 1985).

Educational research is often especially challenging because studies of student growth often involve a doubly nested structure of repeated observations within individuals, who are in turn nested within organizational settings. Research on instruction, for example, focuses on the interactions between students and a teacher around specific curricular materials. These interactions usually occur within a classroom setting and are bounded within a single academic year. The research problem has three foci: the individual growth of students over the course of the academic year (or segment of a year), the effects of personal characteristics and individual educational experiences on student learning, and how these relations are in turn influenced by classroom organization and the specific behavior and characteristics of the teacher. Correspondingly, the data have a three-level hierarchical structure. The level-1 units are the repeated observations over time, which are nested within the level-2 units of persons, who in turn are nested within the level-3 units of classrooms or schools.

Three-level hierarchies also arise frequently in cross-sectional research. For example, Nye, Hedges, and Konstantopoulos (2000) reanalyzed data from the Tennessee class size experiment, which involved students nested within classrooms within schools. Raudenbush, Cheong, and Fotiu (1999) analyzed the U.S. National Assessment of Educational Progress, which involves students nested within schools within states.

Persistent Dilemmas in the Analysis of Hierarchical Data

Despite the prevalence of hierarchical structures in behavioral and social research, past studies have often failed to address them adequately in the data analysis. In large part, this neglect has reflected limitations in conventional statistical techniques for the estimation of linear models with nested structures. In social research, these limitations have generated concerns about aggregation bias, misestimated precision, and the "unit of analysis" problem (see Chapter 5). They have also fostered an impoverished conceptualization, discouraging the formulation of explicit multilevel models with hypotheses about effects occurring at each level and across levels. Similarly, studies of human development have been plagued by "measurement of change" problems (see Chapter 6), which at times have seemed intractable, and have even led some analysts to advise against attempts to directly model growth.

Although these problems of unit of analysis and measuring change have distinct, long-standing, and virtually nonoverlapping literatures, they share a common cause: the inadequacy of traditional statistical techniques for modeling hierarchy. In the past, sophisticated analysts have often been able to find ways to cope at least partially in specific instances. With recent developments in the statistical theory for estimating hierarchical linear models, however, an integrated set of methods now exists that permits efficient estimation for a much wider range of applications.

Even more important from our perspective is that the barriers to use of an explicit hierarchical modeling framework have now been removed. We can now readily pose hypotheses about relations occurring at each level and across levels and also assess the amount of variation at each level. From a substantive perspective, the hierarchical linear model is more homologous with the basic phenomena under study in much behavioral and social research. The applications to date have been encouraging in that they have both afforded an exploration of new questions and provided some empirical results that might otherwise have gone undetected.

A Brief History of the Development of Statistical Theory for Hierarchical Models

The models discussed in this book appear in diverse literatures under a variety of titles. In sociological research, they are often referred to as *multilevel linear models* (cf. Goldstein, 1995; Mason et al., 1983). In biometric applications, the terms *mixed-effects models* and *random-effects models* are common (cf. Elston & Grizzle, 1962; Laird & Ware, 1982; Singer, 1998).

They are also called *random-coefficient regression models* in the econometrics literature (cf. Rosenberg, 1973; Longford, 1993) and in the statistical literature have been referred to as *covariance components models* (cf. Dempster, Rubin, & Tsutakawa, 1981; Longford, 1987).

We have adopted the term *hierarchical linear models* because it conveys an important structural feature of data that is common in a wide variety of applications, including studies of growth, organizational effects, and research synthesis. This term was introduced by Lindley and Smith (1972) and Smith (1973) as part of their seminal contribution on Bayesian estimation of linear models. Within this context, Lindley and Smith elaborated a general framework for nested data with complex error structures.

Unfortunately, the Lindley and Smith (1972) contribution languished for a period of time because use of the models required estimation of covariance components in the presence of unbalanced data. With the exception of some very simple problems, no general estimation approach was feasible in the early 1970s. Dempster, Laird, and Rubin's (1977) development of the EM algorithm provided the needed breakthrough: a conceptually feasible and broadly applicable approach to covariance component estimation. Dempster et al. (1981) demonstrated applicability of this approach to hierarchical data structures. Laird and Ware (1982) and Strenio, Weisberg, and Bryk (1983) applied this approach to the study of growth, and Mason et al. (1983) applied it to cross-sectional data with a multilevel structure.

Subsequently, other numerical approaches to covariance component estimation were also offered through the use of iteratively reweighted generalized least squares (Goldstein, 1986) and a Fisher scoring algorithm (Longford, 1987). A variety of statistical computing programs are available for fitting these models, including HLM (Raudenbush et al., 2000), MIXOR (Hedeker, & Gibbons, 1996), MLWIN (Rasbash et al., 2000), SAS Proc Mixed (Littel et al., 1996), and VARCL (Longford, 1988). Fully Bayesian methods have also been developed by Gelfand et al. (1990) and Seltzer (1993) and are now widely available through software packages such as BUGS (Spiegelhalter et al., 1994).

Early Applications of Hierarchical Linear Models

As noted above, behavioral and social data commonly have a nested structure, including, for example, repeated observations nested within persons. These persons also may be nested within organizational units such as schools. Further, the organizational units themselves may be nested within communities, within states, and even within countries. With hierarchical linear models, each of the levels in this structure is formally represented by its own submodel. These submodels express relationships among variables within a given level, and specify how variables at one level influence relations occurring at another. Although any number of levels can be represented, all the essential statistical features are found in the basic two-level model.

The applications discussed below address three general research purposes: improved estimation of effects within individual units (e.g., developing an improved estimate of a regression model for an individual school by borrowing strength from the fact that similar estimates exist for other schools); the formulation and testing of hypotheses about cross-level effects (e.g., how varying school size might affect the relationship between social class and academic achievement within schools); and the partitioning of variance and covariance components among levels (e.g., decomposing the covariation among set of student-level variables into within- and between-school components). Early published examples of each type are summarized briefly below.

Improved Estimation of Individual Effects

Braun, Jones, Rubin and Thayer (1983) were concerned about the use of standardized test scores for selecting minority applicants to graduate business schools. Many schools base admissions decisions, in part, on equations that use test scores to predict later academic success. However, because most applicants to most business schools are white, their data dominate the estimated prediction equations. As a result, these equations may not produce an adequate ordering for purposes of selecting minority students.

In principle, a separate equation for minority applicants in each school might be fairer, but difficulties often arise in estimating such equations, because most schools have only a few minority students and thus little data on which to develop reliable predictions. In Braun et al.'s (1983) data on 59 graduate business schools, 14 schools had no minorities, and 20 schools had only one to three minority students. Developing prediction equations for minorities in these schools would have been impossible using standard regression methods. Further, even in the 25 schools with sufficient data to sustain a separate estimation, the minority samples were still small, and as a result, the minority coefficients would have been poorly estimated.

Alternatively, the data could be pooled across all schools, ignoring the nesting of students within schools, but this also poses difficulties. Specifically, because minorities are much more likely to be present in some schools than others, a failure to represent these selection artifacts could bias the estimated prediction coefficients.

Braun et al. (1983) used a hierarchical linear model to resolve this dilemma. By borrowing strength from the entire ensemble of data, they were able to efficiently use all of the available information to provide each school with separate prediction equations for whites and minorities. The estimator for each school was actually a weighted composite of the information from that school and the relations that exist in the overall sample. As one might intuitively expect, the relative weights given each component depend on its precision. The estimation procedure is described in Chapter 3 and illustrated in Chapter 4. For a related application, see Rubin (1980), and for a statistical review of these developments, see Morris (1983).

Modeling Cross-Level Effects

The second general use of a hierarchical model is to formulate and test hypotheses about how variables measured at one level affect relations occurring at another. Because such cross-level effects are common in behavioral and social research, the modeling framework provides a significant advance over traditional methods.

An Example from Social Research. Mason et al. (1983) examined the effects of maternal education and urban versus rural residence on fertility in 15 countries. It has been well known that in many countries high levels of education and urban residence predict low fertility. However, the investigators reasoned that such effects might depend on characteristics of the countries, including level of national economic development, as indicated by gross national product (GNP), and the intensity of family planning efforts.

Mason et al. found that higher levels of maternal education were indeed associated with lower fertility rates in all countries. Differences in urban versus rural fertility rates, however, varied across countries with the largest gaps occurring in nations with a high GNP and few organized family planning efforts. The identification of *differentiating effects,* such as this rural-urban gap, and the prediction of their variability is taken up in Chapter 5.

An Example from Developmental Research. Investigators of language development hypothesized that word acquisition depends on two sources: exposure to appropriate speech and innate differences in ability to learn from such exposure. It is widely assumed that innate differences in ability are largely responsible for observed differences in children's vocabulary development. However, the empirical support for this assumption is weak. Heritability studies have found that parent scores on standardized vocabulary tests account for 10% to 20% of the variance in children's scores on the same tests. Exposure studies have not fared much better. Most of the individual variation in vocabulary acquisition has remained unexplained.

In the past, researchers have relied primarily on two-time-point designs to study exposure effects. Children's vocabulary might first be assessed at, say, 14 months of age, and information on maternal verbal ability or language use also might be collected at that time. Children's vocabulary size then would be reassessed at some subsequent time point, say 26 months. The data would be analyzed with a conventional linear model with the focus on estimating the maternal speech effect on 26-month vocabulary size after controlling for children's "initial ability" (i.e., the 14-month vocabulary size).

Huttenlocher et al. (1991) collected longitudinal data with some children observed on as many as seven occasions between 14 and 26 months. This permitted formulation of an individual vocabulary growth trajectory based on the repeated observations for each child. Each child's growth was characterized by a set of parameters. A second model used information about the children, including the child's sex and amount of maternal speech in the home environment, to predict these growth parameters. The actual formulation and estimation of this model are detailed in Chapter 6.

The analysis, based on a hierarchical linear model, found that exposure to language during infancy played a much larger role in vocabulary development than had been reported in previous studies. (In fact, the effects were substantially larger than would have been found had a conventional analysis been performed using just the first and last time points.) This application demonstrates the difficulty associated with drawing valid inferences about correlates of growth from conventional approaches.

Partitioning Variance-Covariance Components

A third use of hierarchical linear models draws on the estimation of variance and covariance components with unbalanced, nested data. For example, educational researchers often wish to study the growth of the individual student learner within the organizational context of classrooms and schools. Formal modeling of such phenomena require use of a three-level model.

Bryk and Raudenbush (1988) illustrated this approach with a small subsample of longitudinal data from the Sustaining Effects Study (Carter, 1984). They used mathematics achievement data from 618 students in 86 schools measured on five occasions between Grade 1 and Grade 3. They began with an individual growth (or repeated measures) model of the academic achievement for each child within each school. The three-level approach enabled a decomposition of the variation in these individual growth trajectories into within- and between-school components. The details of this application are discussed in Chapter 8.

The results were startling—83% of the variance in growth rates was between schools. In contrast, only about 14% of the variance in initial status was between schools, which is consistent with results typically encountered in cross-sectional studies of school effects. This analysis identified substantial differences among schools that conventional models would not have detected, because such analyses do not allow the partitioning of learning-rate variance into within- and between-school components.

New Developments Since the First Edition

As interest in hierarchical models has grown, the pace of methodological innovation has accelerated, with many creative applications in social science and medicine. First, methodologists have expanded the range of outcome variables to which these models apply. Second, models have been extended to include cross-classified as well as purely nested data structures. Third, the applications to multivariate outcome problems have become more prominent. Fourth, latent variable models have been embedded within hierarchical models. Fifth, Bayesian inference for hierarchical models has increased in accessibility and popularity. When combined with the vast increases in computational power now available, this expansion in modeling has inspired a whole array of new applications.

An Expanded Range of Outcome Variables

The first edition of this book confined its attention to continuously distributed outcomes at level 1. For outcomes of this type, it is often reasonable to assume that model errors are normally distributed. Statistical inference proceeds within comparatively tractable normal distribution theory. Inference is much more challenging when outcomes are discrete. Prominent examples include binary outcomes (e.g., employed versus unemployed), counted data (e.g., the number of crimes committed in a given neighborhood per year), ordered categorical outcomes (e.g., low, medium, or high job satisfaction), and multicategory nominal scale outcomes (e.g., occupation type as manual, clerical, service, or professional). For these outcomes, it is implausible to assume linear models and normality at level 1. Embedding such nonnormal outcomes within a hierarchical model makes the task of developing appropriate statistical and computational approaches much more challenging.

Stiratelli, Laird, and Ware (1984) and Wong and Mason (1985) were among the first to tackle these problems using a first-order approximation to maximum likelihood (ML) estimation. Goldstein (1991) and Longford (1993) developed software that made these approaches accessible for several types of discrete outcomes and for two- and three-level models. However, Breslow and

Clayton (1993) and German and Rodriguez (1995) showed that such approximations could be quite inaccurate under certain conditions. Goldstein (1995) developed a second-order approximation. Hedeker and Gibbons (1993) and Pinheiro and Bates (1995) developed accurate approximations to ML using Gauss-Hermite quadrature, and these are now implemented, respectively, in the software packages Mixor and SAS Proc Mixed. An alternative approximation that is typically accurate and computationally convenient uses a high-order Laplace transform (Raudenbush, Yang, & Yosef, 2000) and is implemented in the program HLM.

As a result of these statistical and computational innovations, researchers can now apply the logic of hierarchical models to a vastly expanded array of outcome types. Rumberger (1995) used a two-level model to study the predictors of students' dropping out of school. He identified a number of student-level and school-level risk factors for early drop-out. The level-1 model was a logistic regression model in which the log-odds of early drop-out depended on student background characteristics. Level-1 coefficients varied over schools as a function of school characteristics. Horney, Osgood, and Marshall (1995) provide an interesting application in the context of repeated measures. The aim was to study whether the propensity of high-rate offenders to commit crimes changes with life circumstances such as employment and marital status. The outcome was binary, where each participant either did or did not commit a crime during the prior month. Some explanatory variables also varied with time (e.g., employed versus unemployed), whereas others varied over participants but were invariant with respect to time. The level-1 model was a logistic regression model for individual change in the log-odds of committing crime. Parameters of individual change varied at level 2. Sampson, Raudenbush, and Earls (1997) provide an example with count data as the outcome (the number of homicides per neighborhood), whereas Reardon and Buka (in press) consider a survival model to study the age of onset of substance use of youth nested within neighborhoods.

Chapter 10 considers applications of hierarchical models in the case of binary outcomes, counted data, ordered categories, and multinomial outcomes.

Incorporating Cross-Classified Data Structures

All the examples in the first edition of this book involved "pure" hierarchies. For example, if students are nested within classrooms within schools, we can reasonably assume that each student is a member of one and only one classroom and that each classroom is located in one and only one school. Often, however, nested structures are more complex. Consider, for example, a study of children attending a set of schools and living in a set of neighborhoods. The data do not form a pure hierarchy, because a school will typically draw students from multiple neighborhoods while a neighborhood will also send students to multiple schools. We may view the students as cross-classified by neighborhoods and schools. Raudenbush (1993) developed methodology for this kind of setting. A level-1 model describes the association between student-level predictors and the outcome with each "cell" defined by the cross-classification of neighborhoods and schools. The coefficients of that model can then vary over the neighborhoods and schools. Neighborhood-level and school-level predictors may be of interest and two-way interactions between neighborhood- and school-level characteristics may also be of interest. A second example involves the cross-classification of students by primary school and secondary school attended (Goldstein, 1995).

Crossed classifications can also arise in the context of repeated-measures data. In Chapter 12, we consider the growth of math achievement during the elementary school years. Each year, a student attends a new classroom. In principle, experience in that classroom could "deflect" the student's growth curve in a positive or negative direction. The variance of these deflections is the classroom variance. The data involve the cross-classification of repeated measures by student and classroom. Under certain specified assumptions, the results show that part of the variation in growth curves that would otherwise have been attributable to individual differences among children is instead associated with differences in teacher effectiveness. Taking classroom effects into account also reduces the estimate of temporal instability in student outcomes.

Chapter 12 describes cross-classified random-effects models, illustrating the approach with several examples.

Multivariate Models

The first edition of this book considered univariate outcomes at level 1. Since then, hierarchical models for multivariate outcomes have become more common. Perhaps the most prominent application involves repeated-measures data. Suppose that a longitudinal study uses a fixed data-collection design for every participant. That is, the plan is to collect data on the same set of occasions for every participant. If there are no missing data, such a design will produce a balanced set of data: The number of measures per person will be invariant, as will the time elapsed between occasions. In this setting, classical models for multivariate repeated measures will apply. These approaches make possible a generous array of models for the variances and covariances of the time-series data, including autocorrelated errors and randomly varying slopes, as well as an unrestricted model, which allows a distinct variance for each time point and a distinct covariance for each pair of time points. Jennrich and Schluchter (1986) were among the first to propose methods and software that would apply even in the presence of missing time points. Goldstein (1995) incorporated this idea into a hierarchical model. SAS (1996) implemented a similar approach in its program Proc Mixed as did Hedeker and Gibbons in Mixor (1996) and Raudenbush et al. (2000) in HLM. This second edition considers multivariate growth models in Chapter 6, providing a comparison with results obtained from a standard hierarchical model having a univariate outcome. It also considers extension of this approach to the case in which the repeatedly observed participants are nested in organizations.

Latent Variable Models

A second class of multivariate outcome models becomes extremely useful in missing data problems. Using a variant of Jennrich and Schluchter's (1986) approach, one can estimate the joint distribution of a set of continuous variables from incomplete data. This enables an elegant approach for estimating regressions when predictors are missing at random. The "complete data" are regarded as the observed data augmented by unobserved, latent variables. The observed, incomplete data are used to estimate the associations among these latent variables (cf. Little & Rubin, 1987; Little & Schenker, 1995). The method uses a similar strategy to take into account the measurement error of predictors in estimates of multiple regression models. The first level of the hierarchical model represents associations between the fallible observed data and the latent "true" data. The approach extends nicely to the case in which the fallibly measured variables are characteristics of groups such as schools or neighborhoods. Another natural extension is to estimate the direct and indirect effects of fallibly measured predictors in the context of a hierarchical model. These topics are taken up in Chapter 11.

Item response models, long used in educational testing, are increasingly applied in other domains of social science. Such models represent the probability of a given response as a function of characteristics of the item and "ability" or "latent trait" of the person. The person's trait is the latent variable of interest. Such models can be represented as two-level models where item responses are nested within persons. Chapter 11 also shows how such models can be formulated and then extended to incorporate the nesting of persons within groups.

Bayesian Inference

In the first edition of this book, all statistical inferences were based on the method of maximum likelihood (ML). The ML approach has many desirable properties: Parameter estimates are consistent and asymptotically unbiased and efficient. These estimates have large-sample normal distributions, and statistical tests for comparing models are readily available. These desirable properties are based, however, on large-sample theory. In the case of hierarchical models, the number of higher-level units (e.g., the number of level-2 units in a two-level model) is usually key in determining whether these large-sample properties will apply. If, in fact, the sample size at the highest level is small, statistical inferences may not be trustworthy, depending on the degree to which the data are unbalanced and the particular type of inference sought.

Bayesian methods provide a sensible alternative approach in these cases. Standard errors will tend to be more realistic than under ML. Moreover, by supplying the posterior distribution of each parameter of interest, the Bayesian approach affords a variety of interesting graphical and numerical summaries of evidence about a research question.

The appeal of the Bayesian approach is not new. What is new is the availability of convenient computational methods, particularly in the context of hierarchical data and models. The advances underlying these new approaches involve a family of algorithms using Monte Carlo methods to approximate posterior distributions in settings that were previously regarded as intractable. These methods include data augmentation (Tanner & Wong, 1987) and Gibbs sampling (Gelfand et al., 1990; Gelfand & Smith, 1990). Software implementing these approaches is now available (Spiegelhalter et al., 1996; Rasbash et al., 2000). An interesting example is Seltzer's (1993) reanalysis of the vocabulary data originally published by Huttenlocher et al. (1991) and studied in the first edition of this book. This example involved the vocabulary growth of 22 children during the second year of life. ML inference is at risk in this instance because the number of second-level units (children) is small. The analysis produces the posterior distribution of level-1 coefficients, most notably, the acceleration rates for each child, and of the key level-2 coefficients specifying the association between maternal speech and acceleration. Chapter 13 provides a brief introduction to the logic of Bayesian inference, with applications to hierarchical data.

Organization of the Book

The book is organized into four main sections. The first two sections include Part I, "Logic," and Part II, "Basic Applications." These two sections consist of nine chapters and closely parallel the first edition but with some significant expansions and technical clarifications.

We have added a new section to Chapter 6 on multivariate growth models, expanded the number of illustrative examples throughout Part II, and added further data analytic advice on centering of level-1 predictors. We also introduce some new statistics—plausible value intervals and robust standard errors—not in use when the first edition was published.

Part III, "Advanced Applications," is entirely new. Part IV provides a technical discussion of the methods and computational approaches used in the book.

Part I includes three additional chapters beyond this introductory chapter. Chapter 2 introduces the logic of hierarchical linear models by building on simpler concepts from regression analysis and random-effects analysis of variance. In Chapter 3, we summarize the basic procedures for estimation and inference used with these models. The emphasis is on an intuitive introduction with a minimal level of mathematical sophistication. These procedures are then illustrated in Chapter 4 as we work through the estimation of an increasingly complex set of models with a common data set.

Part II, "Basic Applications," includes five chapters. Chapters 5 and 6 consider the use of two-level models in the study of organizational effects and individual growth, respectively. Chapter 7 discusses research synthesis or meta-analysis applications. This actually represents a special class of applications where the level-1 variances are known. Chapter 8 introduces the three-level model and describes a range of applications, including a key educational research problem—how school organization influences student learning over time. Chapter 9 reviews the basic model assumptions, describes procedures for examining the validity of these assumptions, and discusses what is known about sensitivity of inferences to violation of these various assumptions.

Part III, "Advanced Applications," consists of four chapters. Chapter 10 presents hierarchical models for applications with discrete outcomes. It considers binary outcomes, counts, ordered categorical outcomes, and multicategory nominal outcomes. It provides detailed examples in each case, and considers the distinction between population-average models, also called marginal models (Heagerty & Zeger, 2000), and unit-specific models. Chapter 11 considers latent variable models. Topics of interest include estimating regressions from missing data, estimating regressions when predictors are measured with error, and embedding item response models within the framework of the hierarchical model. Chapter 12 presents cross-classified random effects models. Chapter 13 concerns Bayesian inference for hierarchical linear models.

Part IV provides the statistical theory and computations used throughout the book. It considers univariate linear models with normal level-1 errors, multivariate linear models, and hierarchical generalized linear models.

2 The Logic of Hierarchical Linear Models

- Preliminaries
- A General Model and Simpler Submodels
- Generalizations of the Basic Hierarchical Linear Model
- Choosing the Location of *X* and *W* (*Centering*)
- Summary of Terms and Notation Introduced in This Chapter

This chapter introduces the logic of hierarchical linear models. We begin with a simple example that builds upon the reader's understanding of familiar ideas from regression and analysis of variance (ANOVA). We show how these common statistical models can be viewed as special cases of the hierarchical linear model. The chapter concludes with a summary of some definitions and notation that are used throughout the book.

Preliminaries

A Study of the SES-Achievement Relationship in One School

We begin by considering the relationship between a single student-level predictor variable (say, socioeconomic status [SES]) and one student-level outcome variable (mathematics achievement) within a single, hypothetical school. Figure 2.1 provides a scatterplot of this relationship. The scatter of points is well represented by a straight line with intercept β_0 and slope β_1. Thus, the regression equation for the data is

$$Y_i = \beta_0 + \beta_1 X_i + r_i. \qquad [2.1]$$

The intercept, β_0, is defined as the expected math achievement of a student whose SES is zero. The slope, β_1, is the expected change in math achievement associated with a unit increase in SES. The error term, r_i, rep-

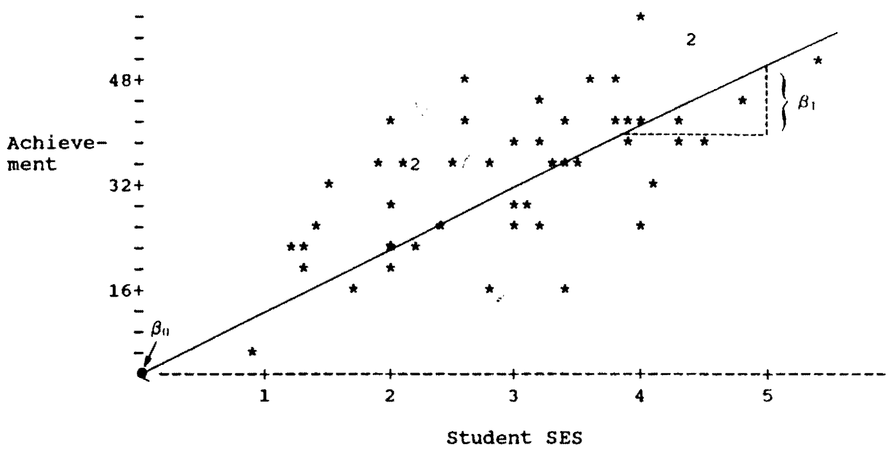

Figure 2.1. Scatterplot Showing the Relationship Between Achievement and SES in One Hypothetical School

resents a unique effect associated with person i. Typically, we assume that r_i is normally distributed with a mean of zero and variance σ^2, that is, $r_i \sim \mathrm{N}(0, \sigma^2)$.

It is often helpful to scale the independent variable, X, so that the intercept will be meaningful. For example, suppose we "center" SES by subtracting the mean SES from each score: $X_i - \overline{X}.$ where $\overline{X}.$ is the mean SES in the school. If we now plot Y_i as a function of $X_i - \overline{X}.$ (see Figure 2.2) with the

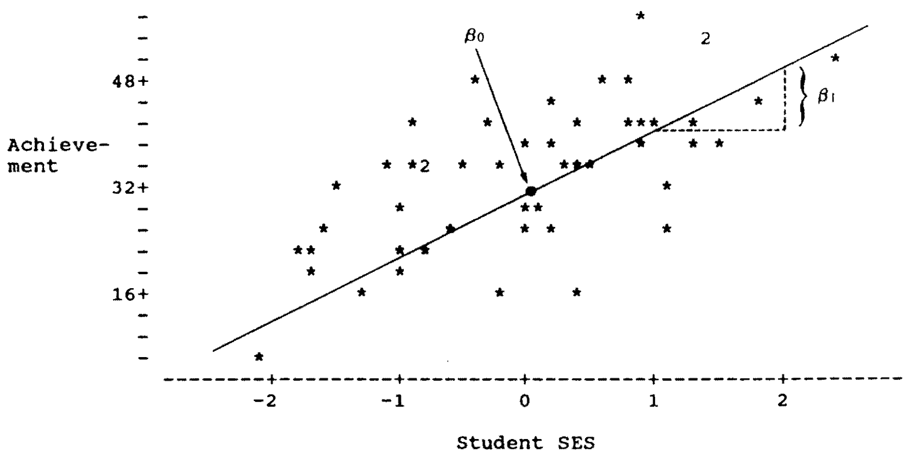

Figure 2.2. Scatterplot Showing the Relationship Between Achievement and SES (Centered) in One Hypothetical School

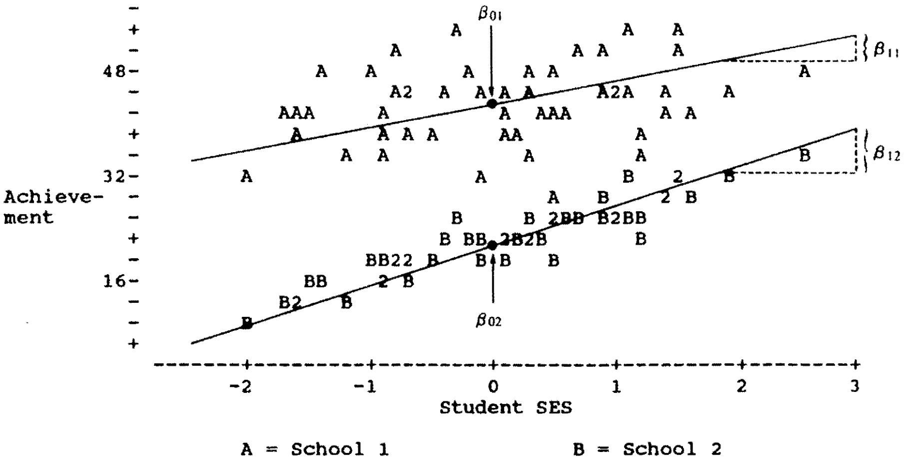

Figure 2.3. Scatterplot Showing the Relationship Between Achievement and SES Within Two Hypothetical Schools

regression line superimposed, we see that the intercept, β_0, is now the mean math achievement while the slope remains unchanged.

A Study of the SES-Achievement Relationship in Two Schools

Let us now consider separate regressions for two hypothetical schools. These are displayed in Figure 2.3. The two lines indicate that School 1 and School 2 differ in two ways. First, School 1 has a higher mean than does School 2. This difference is reflected in the two intercepts, that is, $\beta_{01} > \beta_{02}$. Second, SES is less predictive of math achievement in School 1 than in School 2, as indicated by comparing the two slopes, that is, $\beta_{11} < \beta_{12}$.

If students had been randomly assigned to the two schools, we could say that School 1 is both more "effective" and more "equitable" than School 2. The greater effectiveness is indicated by the higher mean level of achievement in School 1 (i.e., $\beta_{01} > \beta_{02}$). The greater equity is indicated by the weaker slope (i.e., $\beta_{11} < \beta_{12}$). Of course, students are not typically assigned at random to schools, so much interpretations of school effects are unwarranted without taking into account differences in student composition. Nevertheless, the assumption of random assignment clarifies the goals of the analysis and simplifies our presentation.

A Study of the SES-Achievement Relationship in *J* Schools

We now consider the study of the SES-math achievement relationship within an entire *population* of schools. Suppose that we now have a random sample of J schools from a population, where J is a large number. It is no longer practical to summarize the data with a scatterplot for each school. Nevertheless, we can describe this relationship within any school j by the equation

$$Y_{ij} = \beta_{0j} + \beta_{1j}(X_{ij} - \overline{X}_{.j}) + r_{ij}, \qquad [2.2]$$

where for simplicity we assume that r_{ij} is normally distributed with homogeneous variance across schools, that is, $r_{ij} \sim \mathrm{N}(0, \sigma^2)$. Notice that the intercept and slope are now subscripted by j, which allows each school to have a unique intercept and slope. For each school, effectiveness and equity are described by the pair of values (β_{0j}, β_{1j}). It is often sensible and convenient to assume that the intercept and slope have a bivariate normal distribution across the population of schools. Let

$$\mathrm{E}(\beta_{0j}) = \gamma_0, \qquad \mathrm{Var}(\beta_{0j}) = \tau_{00},$$

$$\mathrm{E}(\beta_{1j}) = \gamma_1, \qquad \mathrm{Var}(\beta_{1j}) = \tau_{11},$$

$$\mathrm{Cov}(\beta_{0j}, \beta_{1j}) = \tau_{01},$$

where

γ_0 is the average school mean for the population of schools;
τ_{00} is the population variance among the school means;
γ_1 is the average SES-achievement slope for the population;
τ_{11} is the population variance among the slopes; and
τ_{01} is the population covariance between slopes and intercepts.

A positive value of τ_{01} implies that schools with high means tend also to have positive slopes. Knowledge of these variances and of the covariance leads directly to a formula for calculating the population correlation between the means and slopes:

$$\rho(\beta_{0j}, \beta_{1j}) = \tau_{01}/(\tau_{00}\,\tau_{11})^{1/2}. \qquad [2.3]$$

In reality, we rarely know the true values of the population parameters we have introduced $(\gamma_0, \gamma_1, \tau_{11}, \tau_{00}, \tau_{01})$ nor of the true individual school means and slopes (β_{0j} and β_{1j}). Rather, all of these must be estimated from the data. Our focus in this chapter is simply to clarify the meaning of the parameters. The actual procedures used to estimate them are introduced in Chapter 3 and are discussed more extensively in Chapter 14.

Suppose we did know the true values of the means and slopes for each school. Figure 2.4 provides a scatterplot of the relationship between β_{0j} and

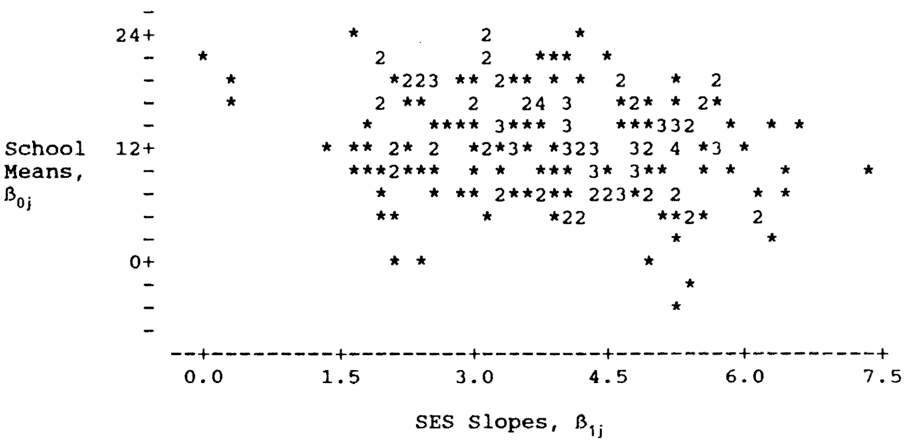

Figure 2.4. Plot of School Means (vertical axis) and SES Slopes (horizontal axis) for 200 Hypothetical Schools

β_{1j} for a hypothetical sample of schools. This plot tells us about how schools vary in terms of their means and slopes. Notice, for example, that there is more dispersion among the means (vertical axis) than the slopes (horizontal axis). Symbolically, this implies that $\tau_{00} > \tau_{11}$. Notice also that the two effects tend to be negatively correlated: Schools with high average achievement, β_{0j}, tend to have weak SES-achievement relationships, β_{1j}. Symbolically, $\tau_{01} < 0$. Schools that are effective and egalitarian—that is, with high average achievement (large values of β_{0j}) and weak SES effects (small values of β_{1j})—are found in the upper left quadrant of the scatterplot.

Having examined graphically how schools vary in terms of their intercepts and slopes, we may wish to develop a model to predict β_{0j} and β_{1j}. Specifically, we could use school characteristics (e.g., levels of funding, organizational features, policies) to predict effectiveness and equity. For instance, consider a simple indicator variable, W_j, which takes on a value of one for Catholic schools and a value of zero for public schools. Coleman, Hoffer, and Kilgore (1982) argued that W_j is positively related to effectiveness (Catholic schools have higher average achievement than do public schools) and negatively related to the slope (SES effects on math achievement are smaller in Catholic than in public schools). We represent these two hypotheses via two regression equations:

$$\beta_{0j} = \gamma_{00} + \gamma_{01} W_j + u_{0j} \qquad [2.4a]$$

and

$$\beta_{1j} = \gamma_{10} + \gamma_{11} W_j + u_{1j}, \qquad [2.4b]$$

where

γ_{00} is the mean achievement for public schools;

γ_{01} is the mean achievement difference between Catholic and public schools (i.e., the Catholic school "effectiveness" advantage);

γ_{10} is the average SES-achievement slope in public schools;

γ_{11} is the mean difference in SES-achievement slopes between Catholic and public schools (i.e., the Catholic school "equity" advantage);

u_{0j} is the unique effect of school j on mean achievement holding W_j constant (or conditioning on W_j); and

u_{1j} is the unique effect of school j on the SES-achievement slope holding W_j constant (or conditioning on W_j).

We assume u_{0j} and u_{1j} are random variables with zero means, variances τ_{00} and τ_{11}, respectively, and covariance τ_{01}. Note these variance-covariance components are now *conditional* or *residual* variance-covariance components. That is, they represent the variability in β_{0j} and β_{1j} remaining after controlling for W_j.

It is not possible to estimate the parameters of these regression equations directly, because the outcomes (β_{0j}, β_{1j}) are not observed. However, the data contain information needed for this estimation. This becomes clear if we substitute Equations 2.4a and 2.4b into Equation 2.2, yielding the single prediction equation for the outcome

$$\begin{aligned} Y_{ij} = \gamma_{00} + \gamma_{01}W_j + \gamma_{10}(X_{ij} - \overline{X}_{\cdot j}) + \gamma_{11}W_j(X_{ij} - \overline{X}_{\cdot j}) \\ + u_{0j} + u_{1j}(X_{ij} - \overline{X}_{\cdot j}) + r_{ij}. \end{aligned} \qquad [2.5]$$

Notice that Equation 2.5 is not the typical linear model assumed in standard ordinary least squares (OLS). Efficient estimation and accurate hypothesis testing based on OLS require that the random errors are independent, normally distributed, and have constant variance. In contrast, the random error in Equation 2.5 is of a more complex form, $u_{0j} + u_{1j}(X_{ij} - \overline{X}_{\cdot j}) + r_{ij}$. Such errors are dependent within each school because the components u_{0j} and u_{1j} are common to every student within school j. The errors also have unequal variances, because $u_{0j} + u_{1j}(X_{ij} - \overline{X}_{\cdot j})$ depend on u_{0j} and u_{1j}, which vary across schools, and on the value of $(X_{ij} - \overline{X}_{\cdot j})$, which varies across students. Though standard regression analysis is inappropriate, such models can be estimated by iterative maximum likelihood procedures described in the

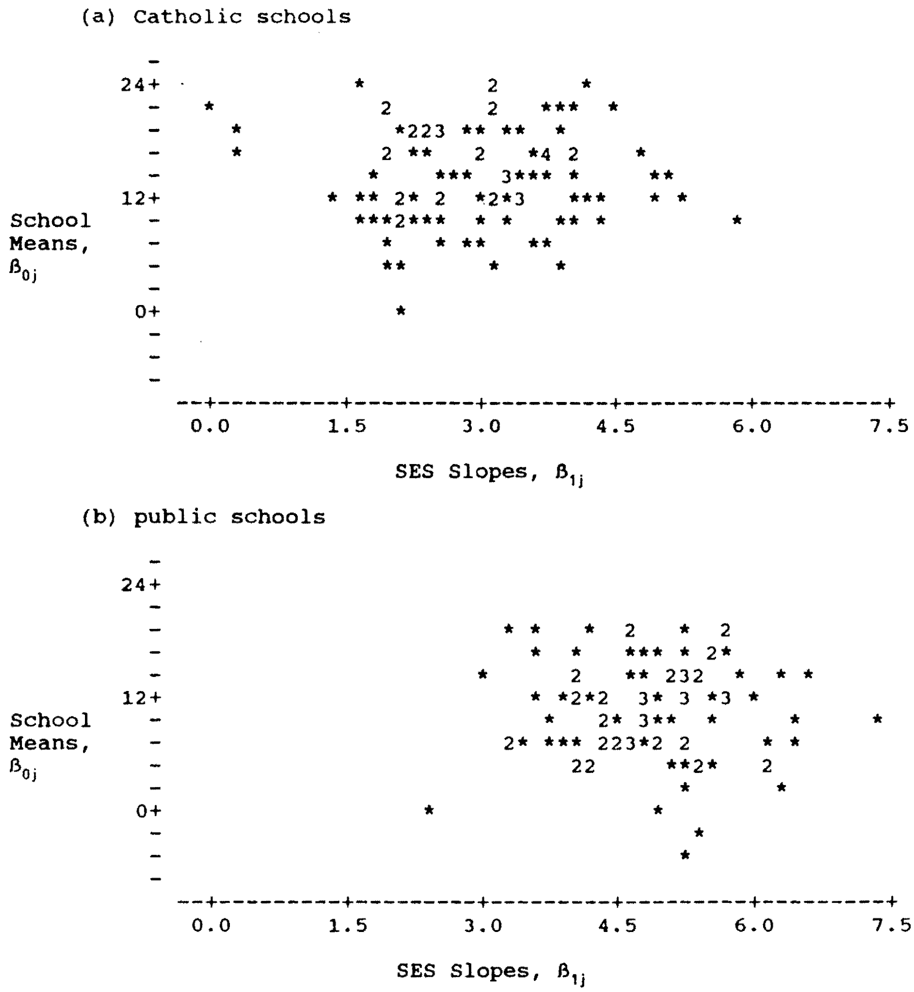

Figure 2.5. Plot of School Means (vertical axis) and SES Slopes (horizontal axis) for 100 Hypothetical Catholic Schools and 100 Hypothetical Public Schools

next chapter. We note that if u_{0j} and u_{1j} were null for every j, Equation 2.5 would be equivalent to an OLS regression model.

Figure 2.5 provides a graphical representation of the model specified in Equation 2.4. Here we see two hypothetical plots of the association between β_{0j} and β_{1j}, one for public and one for Catholic schools. The plots were constructed to reflect Coleman et al.'s (1982) contention that Catholic schools have both higher mean achievement and weaker SES effects than do the public schools.

A General Model and Simpler Submodels

We now generalize our terminology a bit so that it applies to any two-level hierarchical data structure. Equation 2.2 may be labeled the *level-1* model; Equation 2.4 is the *level-2* model, and Equation 2.5 is the *combined* model. In the school-effects application, the level-1 units are students and the level-2 units are schools. The errors r_{ij} are the level-1 random effects and the errors u_{0j} and u_{1j} are level-2 random effects. Moreover, $\mathrm{Var}(r_{ij})$ is the level-1 variance, and $\mathrm{Var}(u_{0j})$, $\mathrm{Var}(u_{1j})$, and $\mathrm{Cov}(u_{0j}, u_{1j})$ are the level-2 variance-covariance components. The β parameters in the level-1 model are level-1 coefficients and the γs are the level-2 coefficients.

Given a single level-1 predictor, X_{ij}, and a single level-2 predictor, W_j, the model given by Equations 2.2, 2.4, and 2.5 is the simplest example of a full hierarchical linear model. When certain sets of terms in this model are set equal to zero, we are left with a set of simpler models, some of which are quite familiar. It is instructive to examine these, both to demonstrate the range of applications of hierarchical linear models and to draw out the connections to more common data analysis methods. The submodels, running from the simpler to the more complex, include the one-way ANOVA model with random effects; a regression model with means-as-outcomes; a one-way analysis of covariance (ANCOVA) model with random effects; a random-coefficients regression model; a model with intercepts- and slopes-as-outcomes; and a model with nonrandomly varying slopes.

One-Way ANOVA with Random Effects

The simplest possible hierarchical linear model is equivalent to a one-way ANOVA with random effects. In this case, β_{1j} in the level-1 model is set to zero for all j, yielding

$$Y_{ij} = \beta_{0j} + r_{ij}. \quad [2.6]$$

We assume that each level-1 error, r_{ij}, is normally distributed with a mean of zero and a constant level-1 variance, σ^2. Notice that this model predicts the outcome within each level-1 unit with just one level-2 parameter, the intercept, β_{0j}. In this case, β_{0j} is just the mean outcome for the jth unit. That is, $\beta_{0j} = \mu_{Y_j}$.

The level-2 model for the one-way ANOVA with random effects is Equation 2.4a with γ_{01} set to zero:

$$\beta_{0j} = \gamma_{00} + u_{0j}, \quad [2.7]$$

where γ_{00} represents the grand-mean outcome in the population, and u_{0j} is the random effect associated with unit j and is assumed to have a mean of zero and variance τ_{00}.

Substituting Equation 2.7 into Equation 2.6 yields the combined model

$$Y_{ij} = \gamma_{00} + u_{0j} + r_{ij}, \qquad [2.8]$$

which is, indeed, the one-way ANOVA model with grand mean γ_{00}; with a group (level-2) effect, u_{0j}; and with a person (level-1) effect, r_{ij}. It is a random-effects model because the group effects are construed as random. Notice that the variance of the outcome is

$$\text{Var}(Y_{ij}) = \text{Var}(u_{0j} + r_{ij}) = \tau_{00} + \sigma^2. \qquad [2.9]$$

Estimating the one-way ANOVA model is often useful as a preliminary step in a hierarchical data analysis. It produces a point estimate and confidence interval for the grand mean, γ_{00}. More important, it provides information about the outcome variability at each of the two levels. The σ^2 parameter represents the within-group variability, and τ_{00} captures the between-group variability. We refer to the hierarchical model of Equations 2.6 and 2.7 as *fully unconditional* in that no predictors are specified at either level 1 or 2.

A useful parameter associated with the one-way random-effects ANOVA is the intraclass correlation coefficient. This coefficient is given by the formula

$$\rho = \tau_{00}/(\tau_{00} + \sigma^2) \qquad [2.10]$$

and measures the proportion of the variance in the outcome that is between the level-2 units. See Chapter 4 for an application of the one-way random-effects submodel.

Means-as-Outcomes Regression

Another common statistical problem involves the means from each of many groups as an outcome to be predicted by group characteristics. This submodel consists of Equation 2.6 as the level-1 model and, for the level-2 model,

$$\beta_{0j} = \gamma_{00} + \gamma_{01} W_j + u_{0j}, \qquad [2.11]$$

where in this simple case we have one level-2 predictor W_j. Substituting Equation 2.11 into Equation 2.6 yields the combined model:

$$Y_{ij} = \gamma_{00} + \gamma_{01} W_j + u_{0j} + r_{ij}. \qquad [2.12]$$

We note that u_{0j} now has a different meaning as contrasted with that in Equation 2.7. Whereas the random variable u_{0j} had been the deviation of unit j's mean from the grand mean, it now represents the residual

$$u_{0j} = \beta_{0j} - \gamma_{00} - \gamma_{01} W_j.$$

Similarly, the variance in u_{0j}, τ_{00}, is now the residual or conditional variance in β_{0j} after controlling for W_j. The advantages of estimating Equation 2.12 rather than performing a standard regression using sample means-as-outcomes are discussed in Chapter 5.

One-Way ANCOVA with Random Effects

Referring again to the full model (Equations 2.2 and 2.4), let us constrain the level-2 coefficients γ_{01} and γ_{11} and the random effects u_{1j} (for all j) equal to 0. The resulting model would be a one-factor ANCOVA with random effects and a single level-1 predictor as a covariate. The level-1 model is Equation 2.2, but now the predictor X_{ij} is centered around the grand mean. That is,

$$Y_{ij} = \beta_{0j} + \beta_{1j}(X_{ij} - \overline{X}_{..}) + r_{ij}. \qquad [2.13]$$

The level-2 model becomes

$$\beta_{0j} = \gamma_{00} + u_{0j}, \qquad [2.14a]$$

$$\beta_{1j} = \gamma_{10}. \qquad [2.14b]$$

Notice that the effect of X_{ij} is constrained to be the same fixed value for each level-2 unit as is indicated by Equation 2.14b.

The combined model becomes

$$Y_{ij} = \gamma_{00} + \gamma_{10}(X_{ij} - \overline{X}_{..}) + u_{0j} + r_{ij}. \qquad [2.15]$$

The only difference between Equation 2.15 and the standard ANCOVA model (cf. Kirk, 1995, chap. 15) is that the group effect here, u_{0j}, is conceived as random rather than fixed. As in ANCOVA, γ_{10} is the pooled within-group regression coefficient of Y_{ij} on X_{ij}. Each β_{0j} is now the mean outcome for each level-2 unit adjusted for differences among these units in X_{ij}. Specifically, $\beta_{0j} = \mu_{Y_j} - \gamma_{10}(\overline{X}_{\cdot j} - \overline{X}_{..})$, where μ_{Y_j} is the mean outcome in school j. We also note that the $\mathrm{Var}(r_{ij}) = \sigma^2$ is now a residual variance after adjusting for the level-1 covariate, X_{ij}.

An extension of the random-effects ANCOVA allows for the introduction of level-2 covariates. For example, if the coefficient γ_{01} is nonnull, the combined model becomes

$$Y_{ij} = \gamma_{00} + \gamma_{01} W_j + \gamma_{10}(X_{ij} - \overline{X}..) + u_{0j} + r_{ij}. \qquad [2.16]$$

This model provides for a level-2 covariate, W_j, while also controlling for the effect of a level-1 covariate, X_{ij}, and the random effects of the level-2 units, u_{0j}. Interestingly, all of the parameters of Equation 2.16 can be estimated using the methods introduced in the next chapter. This is not the case, however, for a classical fixed-effects ANCOVA. Also, the classical ANCOVA model assumes that the covariate effect, γ_{10}, is identical for every group. This homogeneity of regression assumption is easily relaxed using the models described in the next three sections (for randomly varying and nonrandomly varying slopes). We illustrate use of the random-effects ANCOVA model in Chapter 5 in analyzing data on the effectiveness of an instructional innovation on students' writing.

Random-Coefficients Regression Model

All of the submodels discussed above are examples of *random-intercept models*. Only the level-1 intercept coefficient, β_{0j}, was viewed as random. The level-1 slope did not exist in the one-way ANOVA or the means-as-outcomes cases. In the random-effects ANCOVA model, β_{1j} was included but constrained to have a common effect for all groups.

A major class of applications of hierarchical linear models involves studies in which level-1 slopes are conceived as varying randomly over the population of level-2 units. The simplest case of this type is the random-coefficients regression model. In these models, both the level-1 intercept and one or more level-1 slopes vary randomly, but no attempt is made to predict this variation.

Specifically, the level-1 model is identical to Equation 2.2. The level-2 model is still a simplification of Equation 2.4 in that both γ_{01} and γ_{11} are constrained to be null. Hence, the level-2 model becomes

$$\beta_{0j} = \gamma_{00} + u_{0j}, \qquad [2.17a]$$

$$\beta_{1j} = \gamma_{10} + u_{1j}, \qquad [2.17b]$$

where

γ_{00} is the average intercept across the level-2 units;
γ_{10} is the average regression slope across the level-2 units;
u_{0j} is the unique increment to the intercept associated with level-2 unit j; and
u_{1j} is the unique increment to the slope associated with level-2 unit j.

We formally represent the dispersion of the level-2 random effects as a variance-covariance matrix:

$$\text{Var}\begin{bmatrix} u_{0j} \\ u_{1j} \end{bmatrix} = \begin{bmatrix} \tau_{00} & \tau_{01} \\ \tau_{10} & \tau_{11} \end{bmatrix} = \mathbf{T}, \qquad [2.18]$$

where

$\text{Var}(u_{0j}) = \tau_{00}$ = unconditional variance in the level-1 intercepts;
$\text{Var}(u_{1j}) = \tau_{11}$ = unconditional variance in the level-1 slopes; and
$\text{Cov}(u_{0j}, u_{1j}) = \tau_{01}$ = unconditional covariance between the level-1 intercepts and slopes.

Note that we refer to these as unconditional variance-covariance components because no level-2 predictors are included in either Equation 2.17a or 2.17b. Similarly, we refer to Equations 2.17a and 2.17b as an *unconditional* level-2 model.

Substitution of the expressions for β_{0j} and β_{1j} in Equations 2.17a and 2.17b into Equation 2.2 yields a combined model:

$$Y_{ij} = \gamma_{00} + \gamma_{10}(X_{ij} - \overline{X}_{\cdot j}) + u_{0j} + u_{1j}(X_{ij} - \overline{X}_{\cdot j}) + r_{ij}. \qquad [2.19]$$

This model implies that the outcome Y_{ij} is a function of the average regression equation, $\gamma_{00} + \gamma_{10}(X_{ij} - \overline{X}_{\cdot j})$ plus a random error having three components: u_{0j}, the random effect of unit j on the mean; $u_{1j}(X_{ij} - \overline{X}_{\cdot j})$, where u_{1j} is the random effect of unit j on the slope β_{1j}; and the level-1 error, r_{ij}.

Intercepts- and Slopes-as-Outcomes

The random-coefficients regression model allows us to estimate the variability in the regression coefficients (both intercepts and slopes) across the level-2 units. The next logical step is to model this variability. For example, in Chapter 4, we ask "What characteristics of schools (the level-2 units) help predict why some schools have higher means than others and why some schools have greater SES effects than others?"

Given one level-1 predictor, X_{ij}, and one level-2 predictor, W_j, these questions may be addressed by employing the "full model" of Equations 2.2 and 2.4. Of course, this model may be readily expanded to incorporate the effects of multiple Xs and of multiple Ws (see "Generalizations of the Basic Hierarchical Linear Model").

A Model with Nonrandomly Varying Slopes

In some cases, the analyst will prove quite successful in predicting the variability in the regression slopes, β_{1j}. For example, it might be found that the level-2 predictor W_j in Equation 2.4b does indeed predict the level-1 slope β_{1j}. In fact, the analyst might find that after controlling for W_j the residual variance of β_{1j} (i.e., the variance of the residuals, u_{1j} in Equation 2.4b) is very close to zero. The implication would be that once W_j is controlled, little or no variance in the slopes remains to be explained. For reasons of both statistical efficiency and computational stability (as discussed in Chapter 9), it would be sensible, then, to constrain the values of u_{1j} to be zero. This eliminates τ_{11}, the residual variance of the slope, and τ_{01}, the residual covariance between the slope and the intercept, as parameters to be estimated.

If the residuals u_{1j} in Equation 2.4b are indeed set to zero, the level-2 model for the slopes becomes

$$\beta_{1j} = \gamma_{10} + \gamma_{11} W_j, \qquad [2.20]$$

and this model, when combined with Equations 2.2 and 2.4a, yields the combined model

$$\begin{aligned} Y_{ij} = \gamma_{00} + \gamma_{01} W_j + \gamma_{10}(X_{ij} - \overline{X}_{\cdot j}) \\ + \gamma_{11} W_j (X_{ij} - \overline{X}_{\cdot j}) + u_{0j} + r_{ij}. \end{aligned} \qquad [2.21]$$

In this model, the slopes do vary from group to group, but their variation is nonrandom. Specifically, as Equation 2.20 shows, the slopes β_{1j} vary strictly as a function of W_j.

We note that Equation 2.21 can be viewed as another example of what we have called a random-intercept model, because β_{0j} is the only component that varies randomly across level-2 units. In general, hierarchical linear models may involve multiple level-1 predictors where any combination of random, nonrandomly varying, and fixed slopes can be specified.

Section Recap

We have been considering a simple hierarchical linear model with a single level-1 predictor, X_{ij}, and a single level-2 predictor, W_j. In this scenario, the level-1 model (Equation 2.2) defines two parameters, the intercept and the slope. At level 2, each of these may be predicted by W_j and each may have a random component of variation, as in Equations 2.4a and 2.4b. The resulting full model, summarized by Equation 2.5, is the most general model we have considered so far. If certain elements of the full model are constrained to be null, we are left with a submodel that may be useful either as preliminary to a full hierarchical analysis or as a more parsimonious summary than the full model.

The six submodels we have considered may be classified in several different ways. We have distinguished between random-intercept models and randomly varying slope models. The one-way random-effects ANOVA model, the means-as-outcomes model, the one-way ANCOVA model, and the model with nonrandomly varying slopes are all random-intercept models. In such models, the variance components are just the level-1 variance, σ^2, and the level-2 variance, τ_{00}. We noted that in the ANOVA and means-as-outcomes models, no level-1 slope exists. In the ANCOVA model, the level-1 slope exists but is constrained or fixed to be invariant across level-2 units. In the nonrandomly varying slope model, slopes were allowed to vary strictly as a function of a known W_j with no additional random component. In contrast, the random-coefficients model and the slopes- and intercepts-as-outcomes models allowed random variation for both the intercepts and slopes.

Another distinction is whether models include *cross-level interaction terms* such as $\gamma_{11} W_j(X_{ij} - \overline{X}_{\cdot j})$. In general, the combined model will include such cross-level interaction terms whenever we seek to predict variation in a slope. Such terms appear in two of our submodels: the intercepts- and slopes-as-outcomes model and the nonrandomly varying slope model.

Generalizations of the Basic Hierarchical Linear Model

Multiple *X*s and Multiple *W*s

Suppose now that the analyst wishes to use information about a second level-1 predictor. Let X_{1ij} denote the original X discussed above and let X_{2ij} denote the second level-1 predictor. For now, assume that there is still just a single level-2 predictor, W_j. The level-1 model, assuming group-mean centering for both X_{1ij} and X_{2ij}, becomes

$$Y_{ij} = \beta_{0j} + \beta_{1j}(X_{1ij} - \overline{X}_{1\cdot j}) + \beta_{2j}(X_{2ij} - \overline{X}_{2\cdot j}) + r_{ij}. \qquad [2.22]$$

Again, we have three options for modeling β_{2j}. One option is that the effect of X_{2ij} is constrained to be invariant across level-2 units, implying

$$\beta_{2j} = \gamma_{20},$$

where γ_{20} is the common effect of X_{2ij} in every level-2 unit. We say that the effect of β_{2j} is *fixed* across level-2 units.

A second option would be to model the slope β_{2j} as a function of an average value, γ_{20}, plus a random effect associated with each level-2 unit:

$$\beta_{2j} = \gamma_{20} + u_{2j}. \qquad [2.23]$$

Here β_{2j} is *random*. Notice that Equation 2.23 specifies no predictors for β_{2j}. Suppose, however, that this slope depends on W_j. One might then formulate the slopes-as-outcomes model:

$$\beta_{2j} = \gamma_{20} + \gamma_{21} W_j + u_{2j}. \qquad [2.24]$$

According to this model, part of the variation of the slope β_{2j} can be predicted by W_j, but a random component, u_{2j}, remains unexplained. On the other hand, it may be that once the effect of W_j is taken into account, the residual variation in β_{2j}—that is, $\text{Var}(u_{2j}) = \tau_{22}$—is negligible. Then a model constraining that residual variation to be null would be sensible:

$$\beta_{2j} = \gamma_{20} + \gamma_{21} W_j. \qquad [2.25]$$

In this third case, β_{2j} is a *nonrandomly varying* slope because it varies strictly as a function of the predictor W_j.

So far we have been interested in just a single level-2 predictor, W_j. The introduction of multiple W_js is straightforward. Further, the level-2 model does not need to be identical for each equation. One set of W_js may apply for the intercept, a different set be used for β_{1j}, another set for β_{2j}, and so on. When nonparallel specification is employed, however, extra care must be exercised in the interpretation of the results (see Chapter 9).

Generalization of the Error Structures at Level 1 and Level 2

The model specified in Equations 2.2 and 2.4 assumes homogeneous errors at both level 1 and level 2. This assumption is quite acceptable for a broad class of multilevel problems. Most published applications have been based on this assumption, as are most of the examples discussed in Chapter 5 through 8.

The model can easily be extended, however, to more complex error structures at both levels. The level-1 variance might be different for each level-2 unit and denoted σ_j^2, or it might be a function of some measured level-1 characteristic. (The modeling framework for this extension appears in Chapter 5.) Similarly, at level 2, a different covariance structure might exist for distinct subsets of level-2 units. This would result in different T matrices estimated for different subsets of level-2 units.

Extensions Beyond the Basic Two-Level Hierarchical Linear Model

The core ideas introduced in this chapter in the context of two-level models extend directly to models with three or more levels. These extensions are described and illustrated in Chapter 8. A common feature of the basic hierarchical linear model, regardless of the number of levels, is that the outcome variable at level 1, Y, is continuous and assumed normally distributed, conditional on the level-1 predictors included in the model. Over the last decade, extensions beyond the basic hierarchical linear model framework have been advanced to include dichotomous level-1 outcomes, count data, and categorical outcomes. Models for missing data, latent variable effects, and more complex data designs, including crossed random effects, have also appeared. Although the estimation methods are more complex for these extensions, the basic conceptual ideas and modeling framework extend quite naturally. In general, the range of modeling possibilities is now much richer than when we authored the first edition of this book. Part III, which is new to the second edition, introduces these new developments.

Choosing the Location of X and W (*Centering*)

In all quantitative research, it is essential that the variables under study have precise meaning so that statistical results can be related to the theoretical concerns that motivate the research. In the case of hierarchical linear models, the intercept and slopes in the level-1 model become outcome variables at level 2. It is vital that the meaning of these outcome variables be clearly understood.

The meaning of the intercept in the level-1 model depends on the location of the level-1 predictor variables, the Xs. We know, for example, that in the simple model

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + r_{ij}, \tag{2.26}$$

the intercept, β_{0j}, is defined as the expected outcome for a student attending school j who has a value of zero on X_{ij}. If the researcher is to make sense of models that account for variation in β_{0j}, the choice of a metric for all level-1 predictors becomes important. In particular, if an X_{ij} value of zero is not meaningful, then the researcher may want to transform X_{ij}, or "choose a location for X_{ij}" that will render β_{0j} more meaningful. In some cases, a proper choice of location will be required in order to ensure numerical stability in estimating hierarchical linear models.

Similarly, interpretations regarding the intercepts in the level-2 models (i.e., γ_{00} and γ_{10} in Equations 2.4a and 2.4b) depend on the location of the W_j variables. The numerical stability of estimation is not affected by the location for the Ws, but a suitable choice will ease interpretation of results. We describe below some common choices for the location of the Xs and Ws.

Location of the *X*s

We consider four possibilities for the location of X: the natural X metric, centering around the grand mean, centering around the group mean, and other locations for X. We assume that X is measured on an interval scale. The case of dummy variables is considered separately.

The Natural X *Metric.* Although the natural X metric may be quite appropriate in some applications, in others this may lead to nonsensical results. For example, suppose X is a score on the Scholastic Aptitude Test (SAT), which ranges from 200 to 800. Then the intercept, β_{0j}, will be the expected outcome for a student in school j who had an SAT of zero. The β_{0j} parameter is meaningless in this instance because the minimum score on the test is 200. In such cases, the correlation between the intercept and slope will tend toward -1.0. As a result, the intercept is essentially determined by the slope. Schools with strong positive SAT-outcome slopes will tend to have very low intercepts. In contrast, schools where the SAT slope is negligible will tend to have much higher intercepts.

In some applications, of course, an X value of zero will in fact be meaningful. For example, if X is the dosage of an experimental drug, $X_{ij} = 0$ implies that subject i in group j had no exposure to the drug. As a result, the intercept β_{0j} is the expected outcome for such a subject. That is, $\beta_{0j} = \mathrm{E}(Y_{ij}|X_{ij} = 0)$. We wish to emphasize that it is always important to consider the meaning of $X_{ij} = 0$ because it determines the interpretation of β_{0j}.

Centering Around the Grand Mean. It is often useful to center the variable X around the grand mean, as discussed earlier (see "One-Way ANCOVA with Random Effects"). In this case, the level-1 predictors are of the form

$$(X_{ij} - \overline{X}_{..}). \qquad [2.27]$$

Now, the intercept, β_{0j}, is the expected outcome for a subject whose value on X_{ij} is equal to the grand mean, $\overline{X}_{..}$. This is the standard choice of location for X_{ij} in the classical ANCOVA model. As is the case in ANCOVA, grand-mean centering yields an intercept that can be interpreted as an adjusted mean for group j,

$$\beta_{0j} = \mu_{Y_j} - \beta_{1j}(\overline{X}_{\cdot j} - \overline{X}_{..}).$$

Similarly, the $\text{Var}(\beta_{0j}) = \tau_{00}$ is the variance among the level-2 units in the adjusted means.

Centering Around the Level-2 Mean (Group-Mean Centering). Another option is to center the original predictors around their corresponding level-2 unit means:

$$(X_{ij} - \overline{X}_{\cdot j}). \qquad [2.28]$$

In this case, the intercept β_{0j} becomes the unadjusted mean for group j. That is,

$$\beta_{0j} = \mu_{Y_j} \qquad [2.29]$$

and $\text{Var}(\beta_{0j})$ is now just the variance among the level-2 unit means, μ_{Y_j}.

Other Locations for X. Specialized choices of location for X are often sensible. In some cases, the population mean for a predictor may be known and the investigator may wish to define the intercept β_{0j} as the expected outcome in group j for the "average person in the population." In this case, the level-1 predictor would be the original value of X_{ij} minus the population mean.

In applications of two-level hierarchical linear models to the study of growth, the data involve time-series observations so that the level-1 units are occasions and the level-2 units are persons. The investigator may wish to define the metric of the level-1 predictors such that the intercept is the expected outcome for person i at a specific time point of theoretical interest (e.g., entry to school). So long as the data encompass this time point, such a definition is quite appropriate. Examples of this sort are illustrated in Chapters 6 and 8.

Dummy Variables. Consider the familiar level-1 model

$$Y_{ij} = \beta_{0j} + \beta_{1j}X_{ij} + r_{ij}, \qquad [2.30]$$

where X_{ij} is now an indicator or dummy variable. Suppose, for example, that X_{ij} takes on a value of 1 if subject i in school j is a female and 0 if not. In this case, the intercept β_{0j} is defined as the expected outcome for a male student in group j (i.e., the predicted value for student with $X_{ij} = 0$). We note in this case that $\text{Var}(\beta_{0j}) = \tau_{00}$ will be the variance in the male outcome means across schools.

Although it may seem strange at first to center a level-1 dummy variable, this is appropriate and often quite useful. Suppose, for example, that the indicator variable for sex is centered around the grand mean, $\overline{X}_{..}$. This centered predictor can take on two values. If the subject is female, $X_{ij} - \overline{X}_{..}$ will equal the proportion of male students in the sample. If the subject is male, $X_{ij} - \overline{X}_{..}$ will be equal to minus the proportion of female students. As in the case of continuous level-1 predictors centered around the respective grand means, the intercept, β_{0j}, is the adjusted mean outcome in unit j. In this case, it is adjusted for differences among units in the percentage of female students.

Alternatively, we might use group-mean centering. For females, $X_{ij} - \overline{X}_{\cdot j}$ will take on the value equal to the proportion of male students in school j; for males, $X_{ij} - \overline{X}_{\cdot j}$ will take on a value equal to minus the proportion of female students in school j. The fact that X_{ij} is a dummy variable does not change the interpretation given to β_{0j} when group-mean centering is employed. The intercept still represents the average outcome for unit j, μ_{Y_j}.

In sum, several locations of dichotomous predictors will produce meaningful intercepts. Again, it is incumbent on the researcher to take this location into account in interpreting results. Care is especially needed when there are multiple dummy variables. For example, in a school-effects study with indicators for whites, females, and students with preprimary education, the intercept for school j might be the expected outcome for a non-white male student with no preprimary experience. This may or may not be the intercept the investigator wants. Again we offer the general caveat—be conscious of the choice of location for each level-1 predictor because it has implications for interpretation of β_{0j}, $\text{Var}(\beta_{0j})$, and by implication, all of the covariances involving β_{0j}.

In general, sensible choices of location depend on the purposes of the research. No single rule covers all cases. It is important, however, that the researcher carefully consider choices of location in light of those purposes; and it is vital to keep the location in mind while interpreting results.

In addition, the choice of location for the level-1 predictors can, under certain circumstances, also influence the estimation of the level-2 variance-covariance components, **T**, and random level-1 coefficients, β_{qj}. Complications can occur in the context of both organizational research and growth curve applications. The reader is referred to Chapters 5 and 6, respectively, for a further discussion of these technical considerations.

Location of *W*s

In general, the choice of location for the Ws is not as critical as for the level-1 predictors. Problems of numerical instability are less likely, except when cross-product terms are introduced at level 2 (e.g., a predictor set of the form W_{1j}, W_{2j}, and $W_{1j}W_{2j}$). All of the γ coefficients can be easily interpreted whatever choice of metric (or nonchoice) is made for level-2 predictors. Nevertheless, it is often convenient to center all of the level-2 predictors around their corresponding grand means, for example, $W_{1j} - \overline{\overline{W}}_{1\cdot}$.

Summary of Terms and Notation Introduced in This Chapter

A Simple Two-Level Model

Hierarchical form:

$$\begin{aligned} &\text{Level 1 (e.g., students)} & Y_{ij} &= \beta_{0j} + \beta_{1j}X_{ij} + r_{ij}, \\ &\text{Level 2 (e.g., schools)} & \beta_{0j} &= \gamma_{00} + \gamma_{01}W_j + u_{0j}, \\ & & \beta_{1j} &= \gamma_{10} + \gamma_{11}W_j + u_{1j}. \end{aligned}$$

Model in *combined form:*

$$Y_{ij} = \gamma_{00} + \gamma_{10}X_{ij} + \gamma_{01}W_j + \gamma_{11}X_{ij}W_j + u_{0j} + u_{1j}X_{ij} + r_{ij},$$

where we assume:

$$\mathrm{E}(r_{ij}) = 0, \qquad \mathrm{Var}(r_{ij}) = \sigma^2,$$

$$\mathrm{E}\begin{bmatrix} u_{0j} \\ u_{1j} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \mathrm{Var}\begin{bmatrix} u_{0j} \\ u_{1j} \end{bmatrix} = \begin{bmatrix} \tau_{00} & \tau_{01} \\ \tau_{10} & \tau_{11} \end{bmatrix} = \mathbf{T},$$

$$\mathrm{Cov}(u_{0j}, r_{ij}) = \mathrm{Cov}(u_{1j}, r_{ij}) = 0.$$

Notation and Terminology Summary

There are $i = 1, \ldots, n_j$ level-1 units nested with $j = 1, \ldots, J$ level-2 units. We speak of student i nested within school j.

β_{0j}, β_{1j} are level-1 coefficients. These can be of three forms:

fixed level-1 coefficients (e.g., β_{1j} in the one-way random-effects ANCOVA model, Equation 2.14b)

nonrandomly varying level-1 coefficients (e.g., β_{1j} in the nonrandomly-varying-slopes model, Equation 2.20)

random level-1 coefficients (e.g., β_{0j} and β_{1j} in the random-coefficient regression model [Equations 2.17a and 2.17b] and in the intercepts- and slopes-as-outcomes model [Equations 2.4a and 2.4b])

$\gamma_{00}, \ldots, \gamma_{11}$ are level-2 coefficients and are also called fixed effects.

X_{ij} is a level-1 predictor (e.g., student social class, race, and ability).

W_j is a level-2 predictor (e.g., school size, sector, social composition).

r_{ij} is a level-1 random effect.

u_{0j}, u_{1j} are level-2 random effects.

σ^2 is the level-1 variance.

$\tau_{00}, \tau_{01}, \tau_{11}$ are level-2 variance-covariance components.

Some Definitions

Intraclass correlation coefficient (see "One-Way ANOVA with Random Effects"):

$$\rho = \tau_{00}/(\sigma^2 + \tau_{00}).$$

This coefficient measures the proportion of variance in the outcome that is between groups (i.e., the level-2 units). It is also sometimes called the *cluster effect.* It applies only to random-intercept models (i.e., $\tau_{11} = 0$).

Unconditional variance-covariance of β_{0j}, β_{1j} are the values of the level-2 variances and covariances based on the random-coefficient regression model.

Conditional or residual variance-covariance of β_{0j}, β_{1j} are the values of the level-2 variances and covariances after level-2 predictors have been added for β_{0j} and β_{1j} (see, e.g., Equations 2.4a and 2.4b).

Submodel Types

One-way random-effects ANOVA model involves no level-1 or level-2 predictors. We call this a *fully unconditional* model.

Random-intercept model has only one random level-1 coefficient, β_{0j}.

Means-as-outcomes regression model is one form of a random-intercept model.

One-way random-effects ANCOVA model is a classic ANCOVA model, except that the level-2 effects are viewed as random.

Random-coefficients regression model allows all level-1 coefficients to vary randomly. This model is *unconditional at level 2.*

Centering Definitions	**Implications for β_{0j}**
X_{ij} in the natural metric	$\beta_{0j} = \mathrm{E}(Y_{ij} \mid X_{ij} = 0)$
$(X_{ij} - \overline{X}..)$ called grand-mean centering	$\beta_{0j} = \mu_{Y_j} - \beta_{1j}(X_{ij} - \overline{X}..)$ (i.e., adjusted level-2 means)
$(X_{ij} - \overline{X}._j)$ called group-mean centering	$\beta_{0j} = \mu_{Y_j}$ (i.e., level-2 means)
X_{ij} centered at some theoretically chosen location for X	$\beta_{0j} = \mathrm{E}(Y_{ij} \mid X_{ij} =$ chosen centering location for $X)$